Name:_____ Lab Day:_____ CHEM 3B

ANSWER SHEET:

1._____	11._____	21._____	31. _____
2._____	12._____	22._____	
3._____	13._____	23._____	
4._____	14._____	24._____	
5._____	15._____	25._____	
6._____	16._____	26._____	
7._____	17._____	27._____	
8._____	18._____	28._____	
9._____	19._____	29._____	
10._____	20._____	30._____	

PERIODIC TABLE OF THE ELEMENTS

1 1A																	18 8A
1 H 1.008	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (277)	113 (Uut)	114 (Uuq)	115 (Uup)	116 (Uuh)	117 (Uus)	118 (Uuo)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

CHEM 3B Exam 1, Prof Alex Aloia Name:_____ Lab Day_____

MULTIPLE CHOICE. Choose the best answer.

1) What element is most likely to make two covalent bonds? 1) _____
A) Br B) Na C) O D) Ar

2) What is the geometry around the oxygen in the CH_3OCH_3 molecule? 2) _____
A) trigonal planar B) tetrahedral
C) bent D) trigonal pyramidal

3) The bond angle around a carbon atom involved in a double bond is 3) _____
A) 90° B) 120° C) 180° D) 109.5°

4) Which substance will exhibit hydrogen bonding between molecules of itself? 4) _____
A) CH_3CH_2–OH B) $(CH_3)_3N$ C) CH_3–O–CH_3 D) CH_3CH_2–F

5) Which compound would you expect to have the **highest** boiling point? 5) _____
A) dimethyl ether B) methane C) ethanol D) methanol

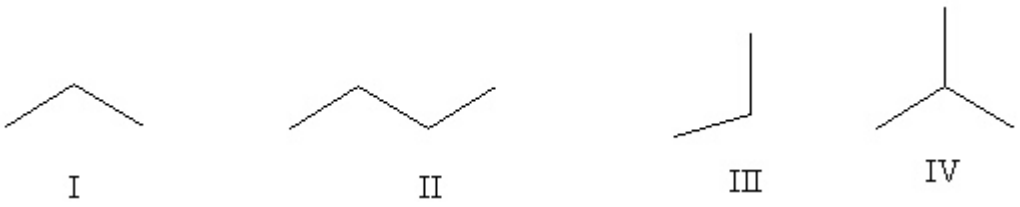
6) Which alcohol is **most** soluble in water? 6) _____
A) 1-butanol B) 1-pentanol C) ethanol D) 1-propanol

7) Which is **not** a property of most alkanes? 7) _____
A) burn readily in an open flame B) soluble in water
C) colorless D) light odor

8) What is a positional isomer of 2-pentanol? 8) _____
A) 2-methyl-2-butanol B) 3-pentanol
C) 3-methyl-2-butanol D) ethyl propyl ether

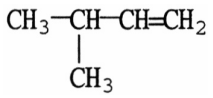
9) Which of the following is a saturated hydrocarbon? 9) _____
A) alcohol B) ether C) alkene D) alkane

10) Which of the following line structures are conformers? 10) _____



A) I and II B) I and III C) II and IV D) None

11) What is the IUPAC name of the molecule shown? 11) _____

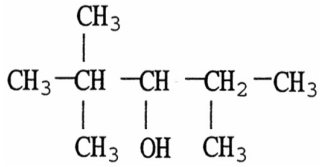


A) 3-methyl-1-butene
B) 1,1-dimethyl-2-propene
C) 2-methyl-3-butene
D) 3-methyl-1,2-butene

12) When an alkyne undergoes a hydration reaction the product is an 12) _____
A) alkene B) alkane C) alcohol D) both A and C

13) What is produced when hydrocarbons undergo complete combustion? 13) _____
A) H_2O and O_2
B) CO_2 and H_2O
C) CO_2 and O_2
D) O_2

14) The name of the alcohol shown is 14) _____

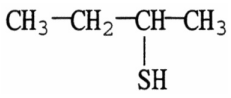


A) trimethyl-3-pentanol
B) 2,2,4-methyl-3-octanol
C) 2,4,4-trimethyl-3-pentanol
D) 2,2,4-trimethyl-3-pentanol

15) Which compound is a tertiary alcohol? 15) _____
A) 2-methyl-1-hexanol
B) 3-hexanol
C) 2-methyl-2-hexanol
D) 3-methyl-2-hexanol

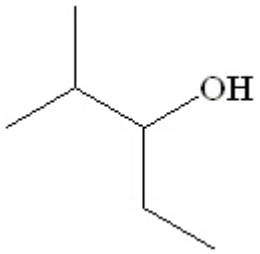
16) When a thiol is oxidized the product is 16) _____
A) a disulfide B) an alkene C) an aldehyde D) a ketone

17) What is the IUPAC name of the compound shown? 17) _____

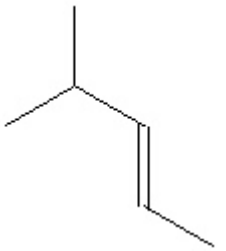


A) 2-thiobutane
B) 1-methyl-1-thiopropane
C) 1-methyl-1-propanethiol
D) 2-butanethiol

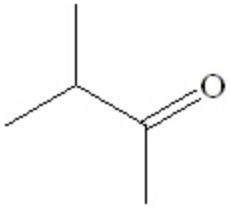
18) Which of the following is the **major** product formed from **dehydration** of the following alcohol? 18) _____



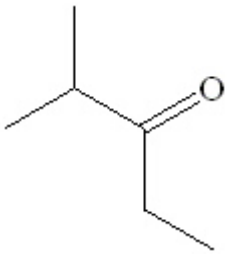
A)



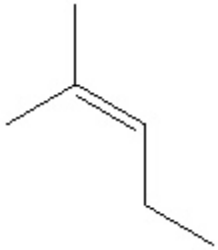
B)



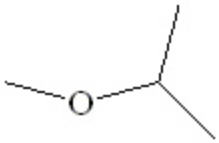
C)



D)



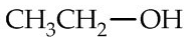
19) What is the **common** name of the following compound? 19) _____



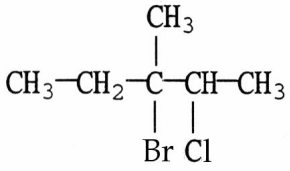
A) ethyl methyl ether
B) 2-methoxy propane
C) methyl isopropyl ether
D) 2-ethoxy propane

20) Which of the following is a tertiary alcohol that cannot be oxidized? 20) _____

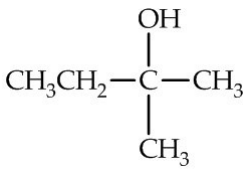
A)



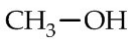
B)



C)



D)

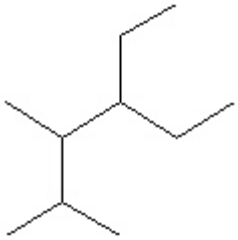


SHORT ANSWER. Write your answer in the space provided.

21) Explain the term "functional group." How do organic chemists use the functional group concept?

22) Write a line structure for 4-chlorooctane.

23) Name the following structure:



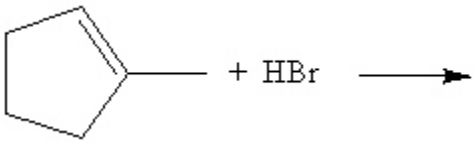
24) A) Draw the structure for 1-pentene

B) Can 1-pentene have *cis-trans* isomers? Why or why not?

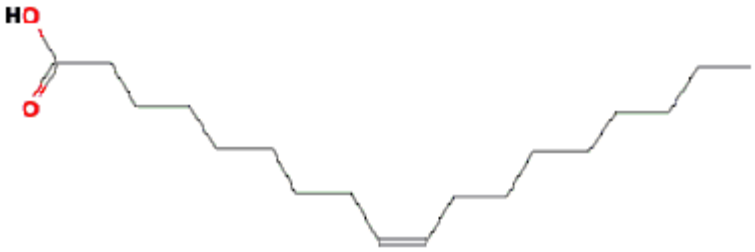
25) A) Draw the structure for p-methytoluene

B) Is p-methyltoluene aromatic? Explain.

26) Predict (draw) the product of the following reaction:

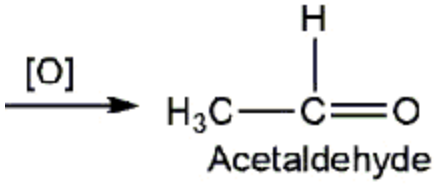


27) A) Is the following stucture a cis or trans isomer?

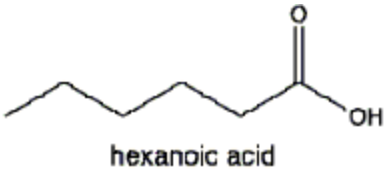


28) What addition reaction can be used to convert an alkyne to an alkene?

29) Draw the structure of the starting material used in this reaction.

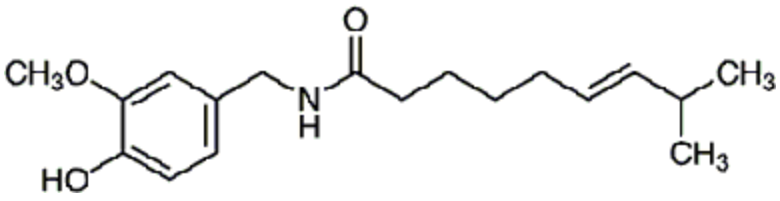


30) Predict (draw) the product when hexanoic acid reacts with a **strong reducing agent**:



31) EXTRA CREDIT:

A) Circle all the functional groups you can identify in the following structure.



B) List the names of the functional groups you identified.